Seed: 65877 Version 3

1. (a) Solve the system: $y = 2x + 2$,
   $3x + y = 20$

   (b) Solve for x and y: $2x + 3y = 15$,
   $4x - y = 8$

2. The determinant of a diagonal matrix equals the sum of its diagonal entries. from Q5

   (A) True

   (B) False

3. If two matrices A and B commute, then $AB = BA$.

   (A) True

   (B) False

4. Answer True or False for each of the following statements:

   (a) The transpose of a symmetric matrix is equal to itself.

      (A) True

      (B) False

   (b) The determinant of a triangular matrix is equal to the product of its diagonal entries.

      (A) True

      (B) False

5. Which of the following represents an antiderivative of e^{2x}?

   A. $\frac{1}{2}e^{2x}$

   B. xe^{2x}

   C. e^{x^2}

6. Donner tous les choix qui sont 1.

   A. 0

B. $[1,\mathrm{un},e^0]$

C. $\frac{\pi}{2}$

7. If $\int_0^a f(x)\,dx = 5$, what is $\int_a^0 f(x)\,dx$?

A. 5

B. -5

C. 0